

Overview

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Part 3 - Training and Development of a Sine-Wave Model

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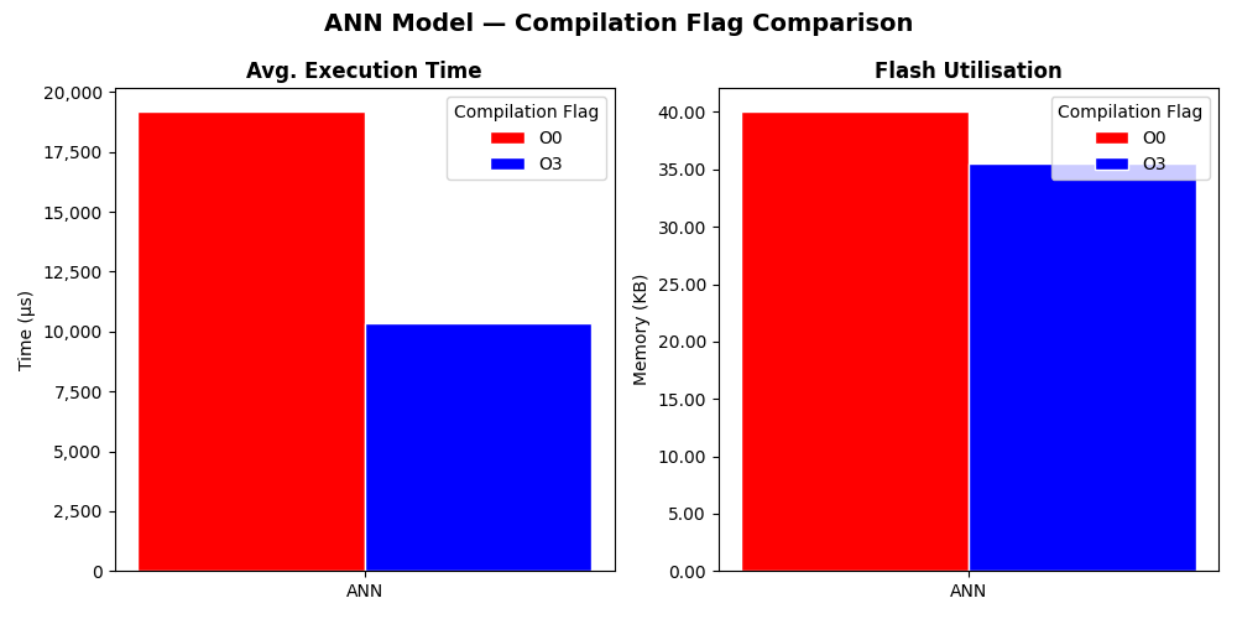
Task 1

Task 2

Task 3

Part 1 - Deployment of ML Algorithms

Model	Compilation Flag	Av. Inference Time (µs)	Memory Utilisation (KB)	
			RAM	Flash
ANN	O0	19196	3.80	40.06
	O3	10362	3.80	35.54



- Execution time is measured using ticks which are found in the low-level hardware abstraction layer.
- The number of runs used to obtain average is determined by `max_run`.
- Double to static const double so that the variable is stored in flash rather than RAM.

Part 2 - Deployment and Performance of ML Algorithms

ResNet

Model	Processor	Optimisation	Inference Time (µs)	Memory Utilisation (KB)		Accuracy (%)
				RAM	Flash	
ST Edge AI Model Zoo	M33 (STM32H573i-DK @ 250 MHz)	N/A	85310	51.01	134.48	87.01
		Time	79940	64.18	134.45	87.01
	M7	N/A	52884	67.88	242.28	N/A
		Time	52884	67.88	242.28	N/A

History of benchmark results

Show performance summary

Export as CSV

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☒	Date ↓	Model Name	Model Type	Version	Optimization	Use Neural-Art Accelerator™	Platform	MACC	Duration (ms)	Flash size	RAM size	Dev
☒	5/1/2026, 4:25:51 PM	resnet8_32_tfs_int8	tfllite	8.1.0	time	false	STM32 MCU	12541060	79.94	131.3 KiB	62.68 KiB	STI DK
☒	5/1/2026, 4:25:02 PM	resnet8_32_tfs_int8	tfllite	8.1.0	ram	false	STM32 MCU	12541060	82.53	134.74 KiB	44.87 KiB	STI DK
☒	5/1/2026, 4:23:22 PM	resnet8_32_tfs_int8	tfllite	8.1.0	balanced	false	STM32 MCU	12541060	85.31	131.33 KiB	49.81 KiB	STI DK

- RAM usage was increased so that the inference time could decrease.

Part 2 - Deployment and Performance of ML Algorithms

Binarized Neural Network

Model	Processor	Optimisation	Inference Time (μ s)	Memory Utilisation (KB)		Accuracy (%)
				RAM	Flash	
BNN on Cifar-10	M33 (STM32H573i-DK @ 250 MHz)	N/A	314900	149.97	556.89	100
		Time	314900	149.97	556.89	100
	M7	N/A	255818	151.77	725.84	N/A
		Time	255895	151.77	722.23	N/A



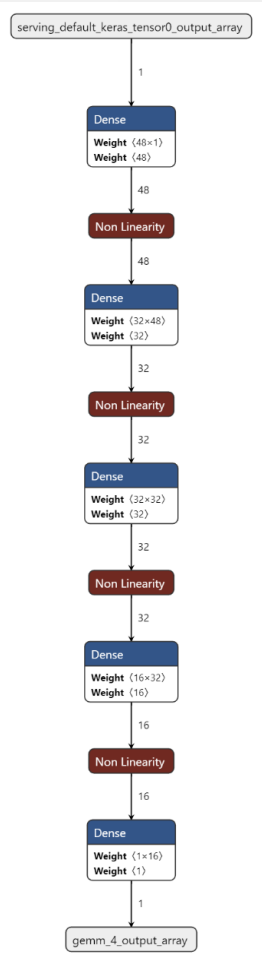
Part 3 - Training and Development of a Sine-Wave Model

Summary of Results

Deployment	Processor	Optimisation	Av. Inference Time (μ s)	Memory Utilisation (KB)		Evaluation Metrics		
				RAM	Flash	MAE	MSE	RMSE
Case 1: CubeIDE	M7	N/A	234	20.48	107.59	0.0562	0.0043	0.0659
Case 2: CubeAI	M7	N/A	89	8.14	48.16	0.0266	0.0012	0.0347
Case 3: ST Edge AI	M7	Time	85	12.04	55.08	0.0562	0.0043	0.0659

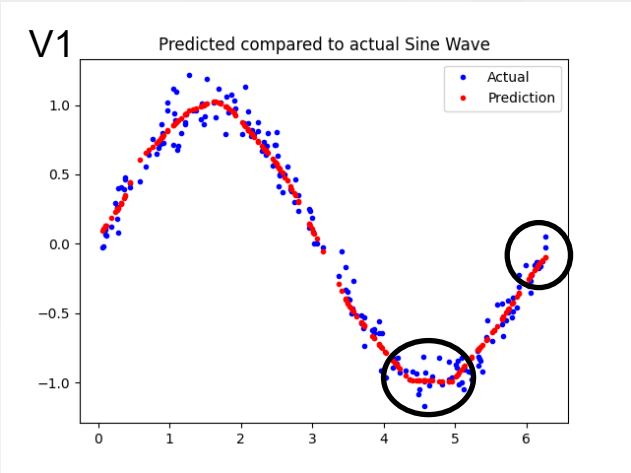
Part 3 - Training and Development of a Sine-Wave Model

Case 1

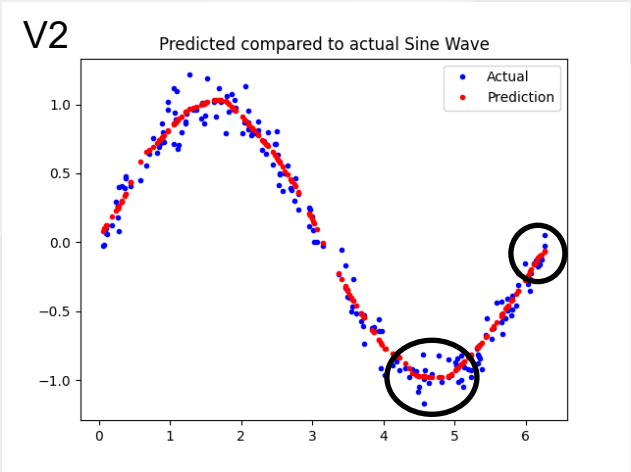


Deployment	Processor	Optimisation	Av. Inference Time (μ s)	Memory Utilisation (KB)		Evaluation Metrics		
				RAM	Flash	MAE	MSE	RMSE
Case 1: CubeIDE	M7	N/A	234	20.48	107.59	0.0562	0.0043	0.0659

Total trainable parameters: 1393



V2



Total trainable parameters: 3265

Part 3 - Training and Development of a Sine-Wave Model

Case 2

COM7 - PuTTY

V1

```
Test case 10: Input = 3.7499 rad (216.00 deg)
Output = [-0.622125 -0.587785 +0.034340] > Negative: Red LED ON
Inference time: 60 us

Test case 11: Input = 4.1888 rad (240.00 deg)
Output = [-0.876465 -0.866025 +0.010440] > Negative: Red LED ON
Inference time: 60 us

Test case 12: Input = 4.6077 rad (264.00 deg)
Output = [-0.998113 -0.994322 -0.000099] > Negative: Red LED ON
Inference time: 60 us

Test case 13: Input = 5.0265 rad (288.00 deg)
Output = [-0.956376 -0.951056 +0.005320] > Negative: Red LED ON
Inference time: 60 us

Test case 14: Input = 5.4454 rad (312.00 deg)
Output = [-0.496870 -0.743145 -0.044879] > Negative: Red LED ON
Inference time: 60 us

Test case 15: Input = 5.8643 rad (336.00 deg)
Output = [-0.382001 -0.406736 -0.024735] > Negative: Red LED ON
Inference time: 60 us

Test case 16: Input = 6.2832 rad (360.00 deg)
Output = [-0.054234 0.000000 +0.054235] > Intersecting Zero: Both LEDs OFF
Inference time: 60 us

Average time : 61 us
MAE (Mean Absolute Error): 0.028237
MSE (Mean Squared Error): 0.001215
RMSE (Root Mean Squared Error): 0.034855

16 Inference tests complete.
```

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

V2

```
Test case 15: Input = 5.8643 rad (336.00 deg)
Output = [-0.376995 -0.406736 -0.023742] > Negative: Red LED ON
Inference time: 87 us

Test case 16: Input = 6.2832 rad (360.00 deg)
Output = [-0.060778 0.000000 +0.060779] > Intersecting Zero: Both LEDs OFF
Inference time: 87 us

Average time : 89 us
MAE (Mean Absolute Error): 0.026594
MSE (Mean Squared Error): 0.001196
RMSE (Root Mean Squared Error): 0.033711

16 Inference tests complete.
```

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

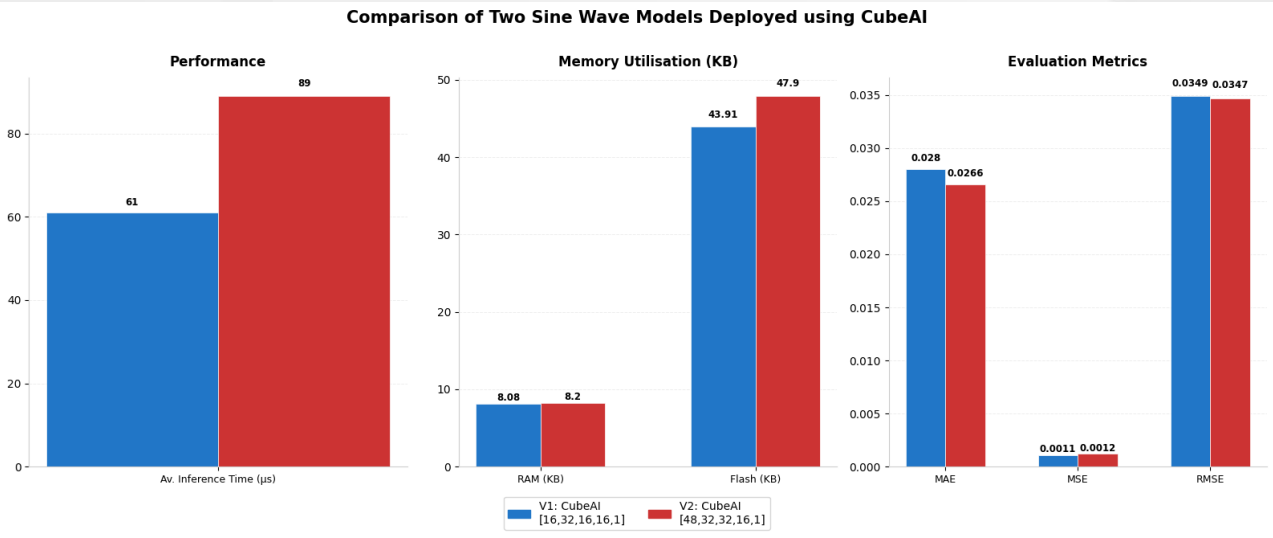
Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

Negative: Red LED ON

Deployment	Processor	Compiler Flag	Av. Inference Time (µs)	Memory Utilisation (KB)		Evaluation Metrics		
				RAM	Flash	MAE	MSE	RMSE
V1: CubeAI [16,32,16,16,1]	M7	O3	61	8.08	43.91	0.028	0.0011	0.0349
V2: CubeAI [48,32,32,16,1]	M7	O3	89	8.2	47.90	0.0266	0.0012	0.0347



Part 3 - Training and Development of a Sine-Wave Model

Case 3

```
COM4 - PuTTY

Test case 14: Input = 4.2990 rad (246.32 deg)
Output = [-0.922653 -0.915773 +0.006880] > Negative: Red LED ON
Inference time: 84 us

Test case 15: Input = 4.6297 rad (265.26 deg)
Output = [-1.029639 -0.996585 +0.033055] > Negative: Red LED ON
Inference time: 83 us

Test case 16: Input = 4.9604 rad (284.21 deg)
Output = [-1.043906 -0.969400 +0.074506] > Negative: Red LED ON
Inference time: 83 us

Test case 17: Input = 5.2911 rad (303.16 deg)
Output = [-0.895671 -0.837166 +0.058505] > Negative: Red LED ON
Inference time: 83 us

Test case 18: Input = 5.6218 rad (322.11 deg)
Output = [-0.686783 -0.614213 +0.072570] > Negative: Red LED ON
Inference time: 83 us

Test case 19: Input = 5.9525 rad (341.05 deg)
Output = [-0.442421 -0.324700 +0.117721] > Negative: Red LED ON
Inference time: 83 us

Test case 20: Input = 6.2832 rad (360.00 deg)
Output = [-0.135077 0.000000 +0.135077] > Negative: Red LED ON
Inference time: 83 us

Average time : 84 us

MAE (Mean Absolute Error): 0.056243
MSE (Mean Squared Error): 0.004340
RMSE (Root Mean Squared Error): 0.065882

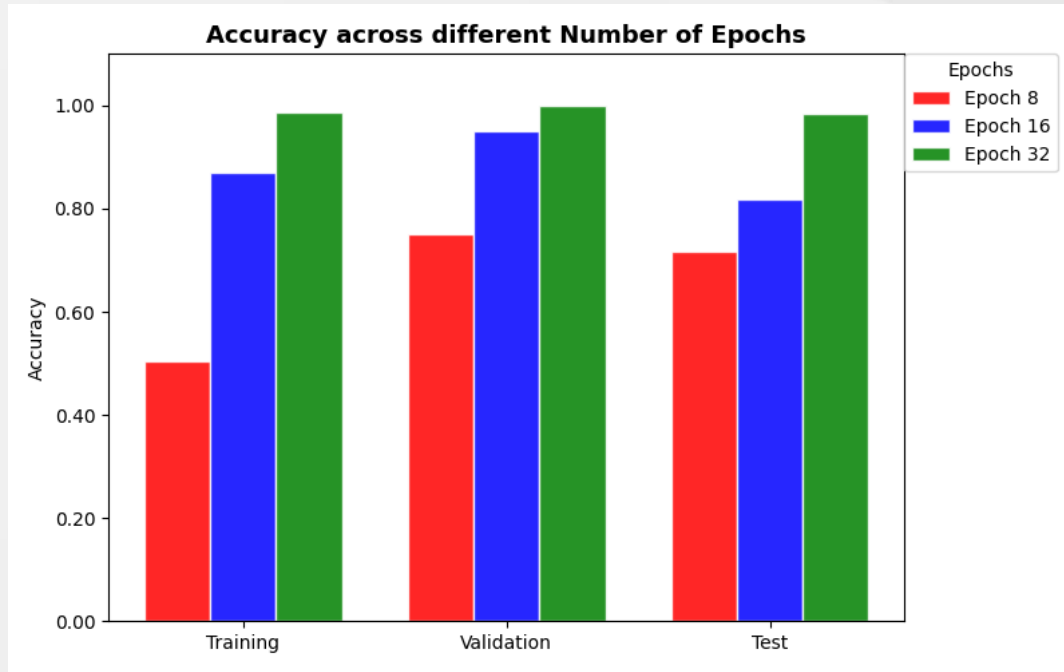
20 Inference tests complete
```

Deployment	Processor	Optimisation	Av. Inference Time (µs)	Memory Utilisation (KB)		Evaluation Metrics		
				RAM	Flash			
						MAE	MSE	RMSE
Case 3: ST Edge AI	M7	Time	84	12.04	55.08	0.0562	0.0043	0.0659



Part 4 - Face Classification using CNNs

Task 1



- At 8 epochs low confidence predictions were still being made.
- This suggested the model was underfitting.
- More epochs means more complete passes through the training dataset

```
(.venv) PS E:\STM32Projects\CW_Part_4> python infer.py --image data/test/circlepolis/circlepolis_190.png
Prediction: circlepolis
Confidence: 0.574
Warning: low confidence :-(. The image may belong to a face type that was not represented in training.
circlepolis 0.574
rectanglepolis 0.426

Training scope:
trained classes : ['circlepolis', 'rectanglepolis']
```

Part 4 - Face Classification using CNNs

Task 2

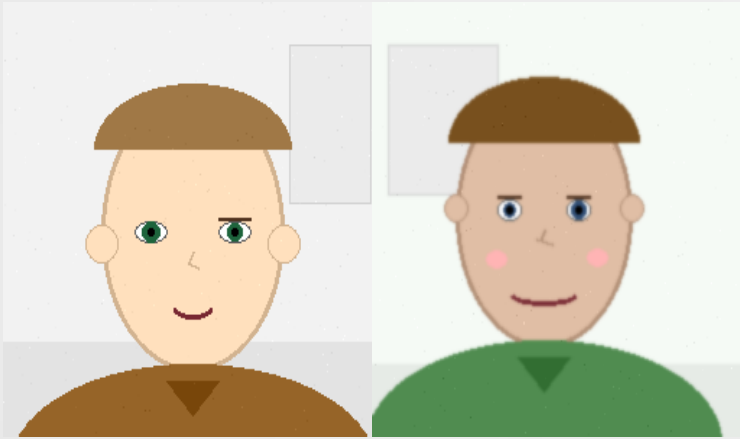
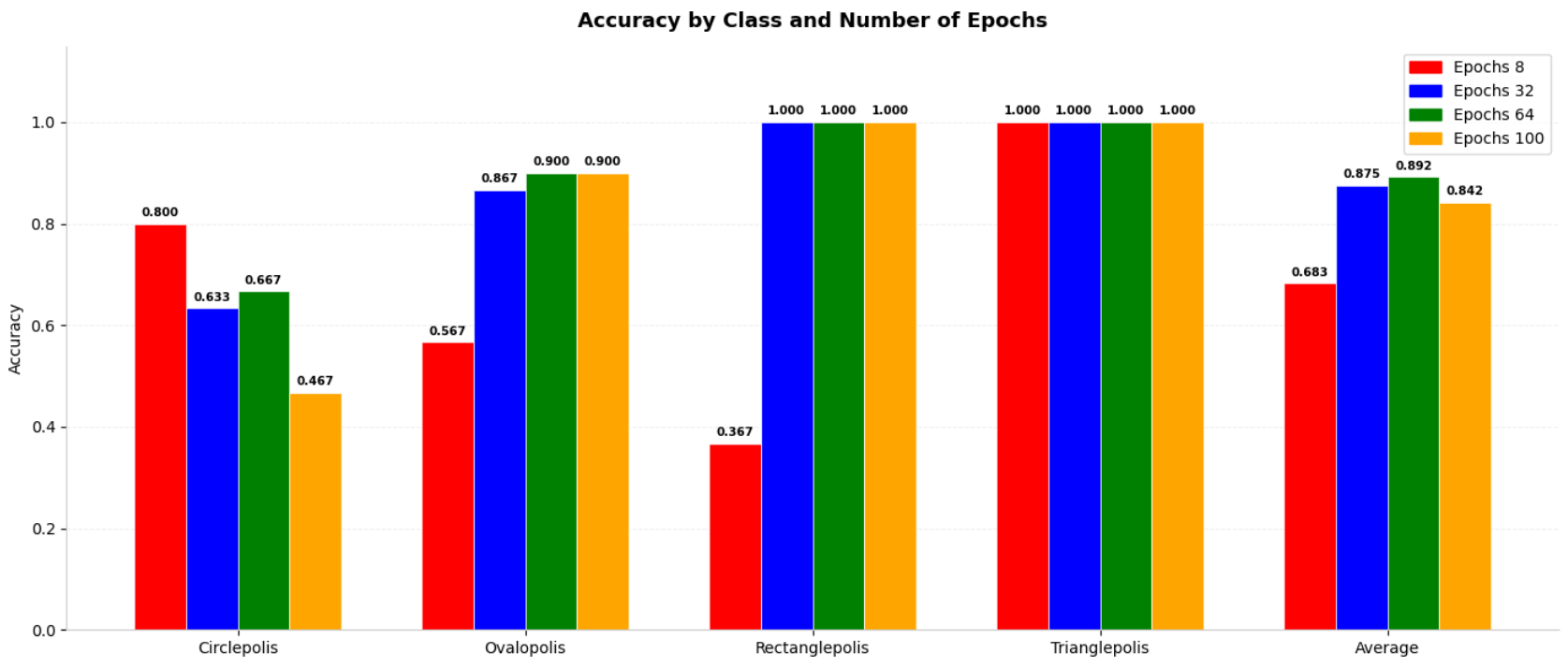
Model Type	Classes	Accuracy	
		Number of Epochs	
		32	64
Two Class	Circlepolis	0.900	0.933
	Rectanglepolis	0.986	0.900
Four Class	Circlepolis	1.000	0.933
	Ovalopolis	-	-
	Rectanglepolis	0.967	0.900
	Trianglepolis	-	-

Test Image	Predicted Class	Accuracy	Was the correct prediction made?
circlepolis_190	Circlepolis	0.999	Y
ovalopolis_185	Ovalopolis	0.986	N
rectanglepolis_176	Rectanglepolis	0.997	Y
trianglepolis_170	Trianglepolis	0.870	N

- Unseen classes are misclassified.
- Model uncertainty shows it has not just memorised the training data (overfitting).

Part 4 - Face Classification using CNNs

Task 3



Class	Accuracy based on epochs		Was the correct prediction made on 32 epochs?
	32	64	
Circlepolis	0.991	0.973	Y
Ovalopolis	0.613	0.870	N
Rectanglepolis	0.997	1.000	Y
Trianglepolis	1.000	1.000	Y